Royal Crest Academy
SCIENCE · INNOVATION · SKILL

FREE LEARNER GUIDE

Biology *study guide*

From a single cell to a whole ecosystem, Biology explains how living things are built, how they work and how they depend on each other.

Big ideas

The cell is the unit of life

Every living thing is made of cells. Plant and animal cells both have a nucleus, cytoplasm and a cell membrane; plant cells also have a cell wall, chloroplasts and a large vacuole.

Energy flows, matter cycles

Plants capture light energy in photosynthesis. Energy passes along food chains and is lost as heat, while carbon, nitrogen and water are recycled again and again.

Variation and inheritance

Genes on chromosomes pass from parents to offspring. Variation, together with natural selection, explains how populations change over time.

Topic checklist

tick each topic when you can explain it

O LEVEL

- ☐ Cell structure and organisation
- ☐ Movement in and out of cells
- ☐ Biological molecules and enzymes
- ☐ Plant nutrition and photosynthesis
- ☐ Human nutrition and digestion
- ☐ Transport in plants and animals
- ☐ Respiration and gas exchange
- ☐ Excretion and homeostasis
- ☐ Coordination and response
- ☐ Reproduction
- ☐ Inheritance and variation
- ☐ Organisms and their environment

A LEVEL

- ☐ Cell structure and membranes
- ☐ Biological molecules and enzymes
- ☐ Cell division and the genetic code
- ☐ Transport and gas exchange
- ☐ Energy and respiration
- ☐ Photosynthesis
- ☐ Homeostasis and control
- ☐ Genetics and gene technology
- ☐ Classification, biodiversity and conservation
- ☐ Evolution and natural selection

Key terms

Diffusion	The net movement of particles from a region of higher concentration to a region of lower concentration.
Osmosis	The movement of water molecules from a dilute solution to a more concentrated one, through a partially permeable membrane.
Enzyme	A protein that speeds up a chemical reaction in a living organism without being used up.
Photosynthesis	Green plants make glucose from carbon dioxide and water using light energy, releasing oxygen.
Respiration	The release of energy from glucose, which happens in every living cell.
Homeostasis	Keeping conditions inside the body, such as temperature and water content, within set limits.
Gene	A length of DNA that codes for a protein, and so controls a characteristic.
Ecosystem	A community of living organisms together with their physical environment.

In the laboratory

- Using a light microscope and drawing what you see
- Food tests: iodine for starch, Benedict's for reducing sugar, biuret for protein
- Testing a leaf for starch
- Investigating osmosis with potato strips
- Measuring how temperature affects an enzyme
- Fieldwork with quadrats and keys

Study smarter

1. Draw and label diagrams from memory, then check them.
2. Learn definitions word for word: examiners look for key words.
3. Always link a structure to its function.
4. Practise reading graphs and tables from experiments.

Check yourself

1. **Name three structures found in plant cells but not in animal cells.**

2. **Why does a potato strip placed in a concentrated salt solution get shorter?**

3. **Write the word equation for photosynthesis.**

Answers 1. A cell wall, chloroplasts and a large permanent vacuole. · 2. Water leaves the potato cells by osmosis into the more concentrated solution, so the cells lose water and shrink. · 3. carbon dioxide + water → glucose + oxygen (in the presence of light and chlorophyll).

Where it leads: Medicine, Nursing, Pharmacy, Agriculture, Veterinary science, Environmental science, Biotechnology